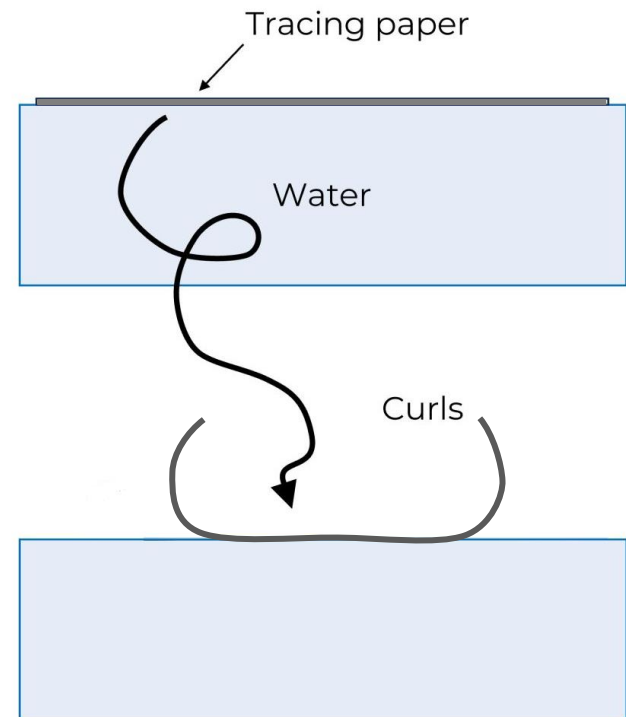


Wet scroll

Gently place a piece of **tracing paper** on the surface of **water**. It **rapidly curls** into a scroll and then **slowly uncurls**.

Explain and investigate this phenomenon.



INTRODUCTION

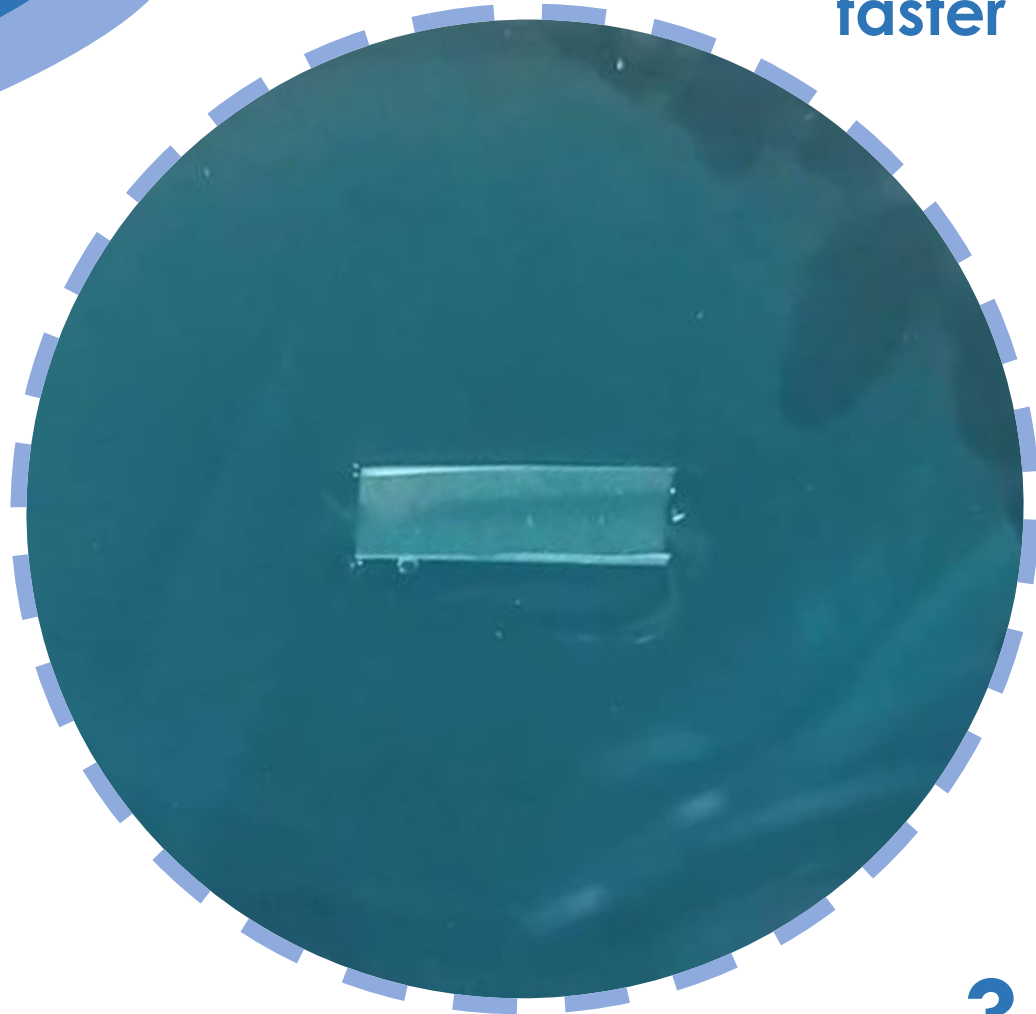
Phenomenon demonstration

Phenomenon demonstration

short paper

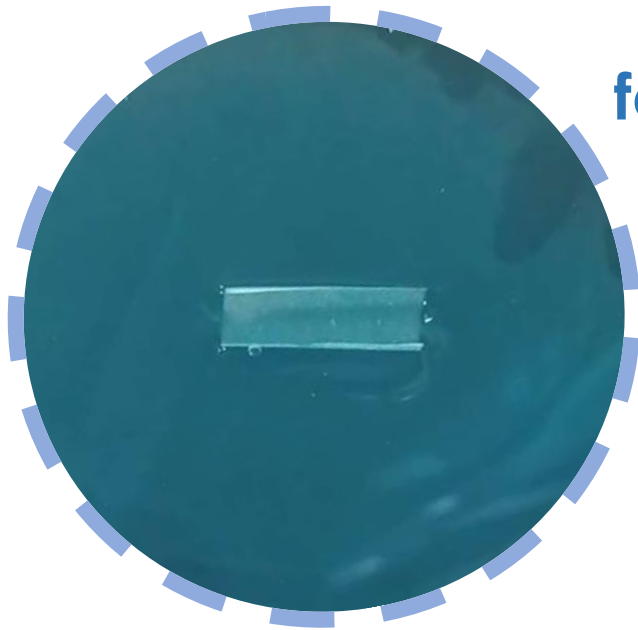
8x
faster

paper curls and
slowly uncurls



water pours in

8x
faster

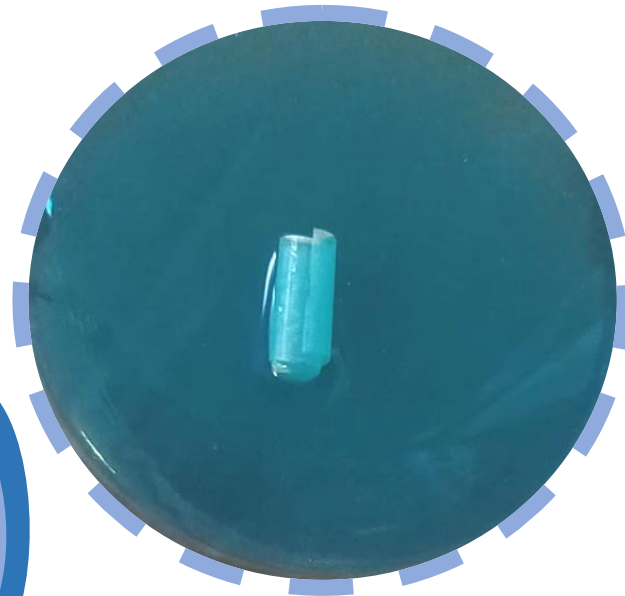


paper curls and
swiftly uncurls

paper curls and
does not uncurl

Other
phenomena

long paper



EXPLANATION

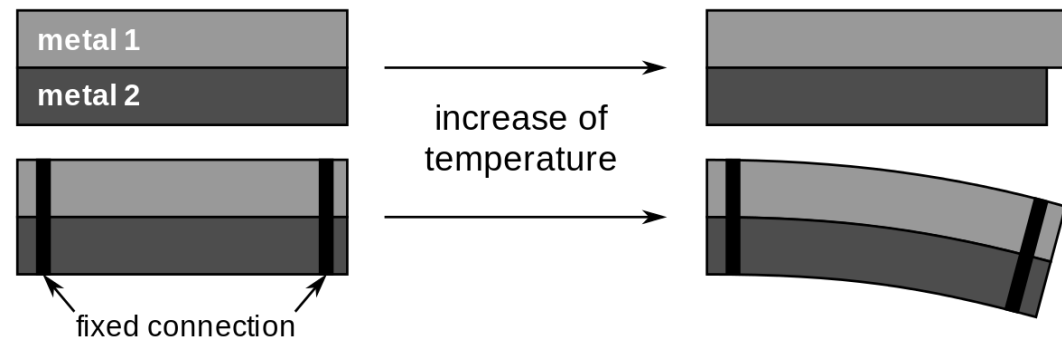
What causes the phenomenon?

Bimetallic strip analogy

How does it work?



$$C_{te}(M1) > C_{te}(M2)$$



Paper strip

Paper curls



Paper strip

And slowly uncurls



Wetting of paper layers

Bottom layer gets wet

dry layer

wet layer

Difference in wetness

water

Wetting of paper layers

Expansion of bottom layers

Swelling of wet paper



Wetting of paper layers

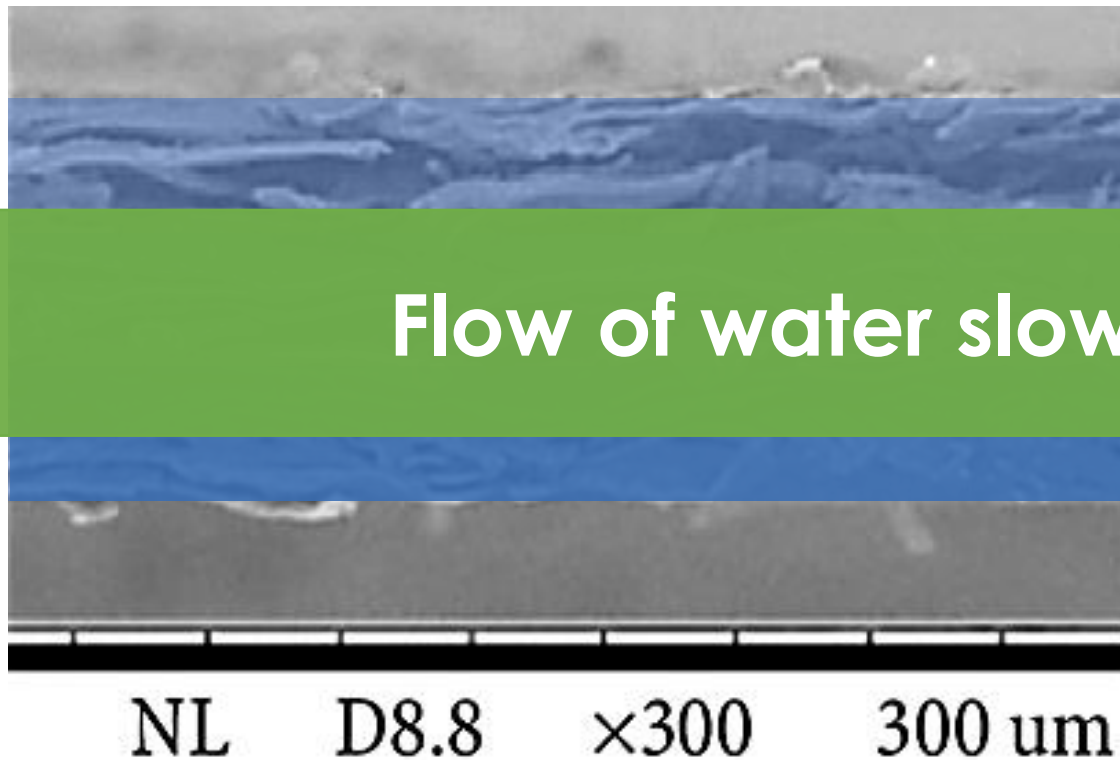
Balancing wetness

Whole paper is wet -» uncurling

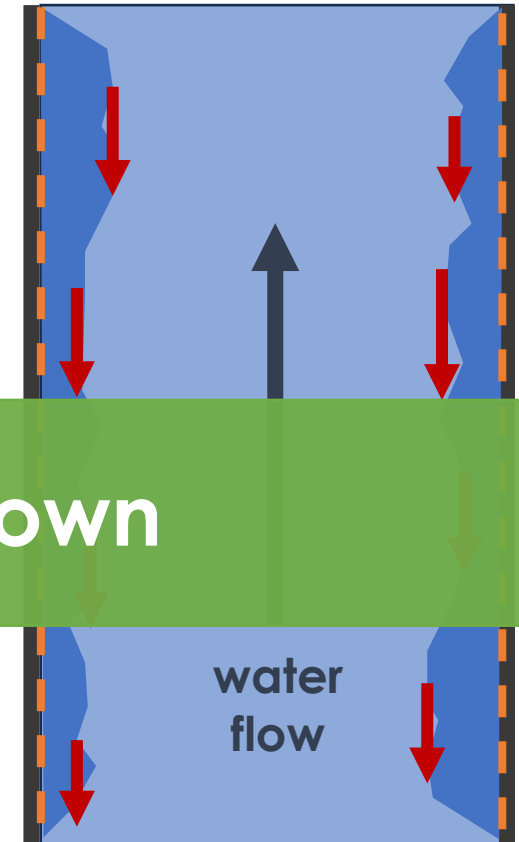


Why it *curls rapidly* and *uncurls slowly*?

Viscosity effect - Hagen–Poiseuille law



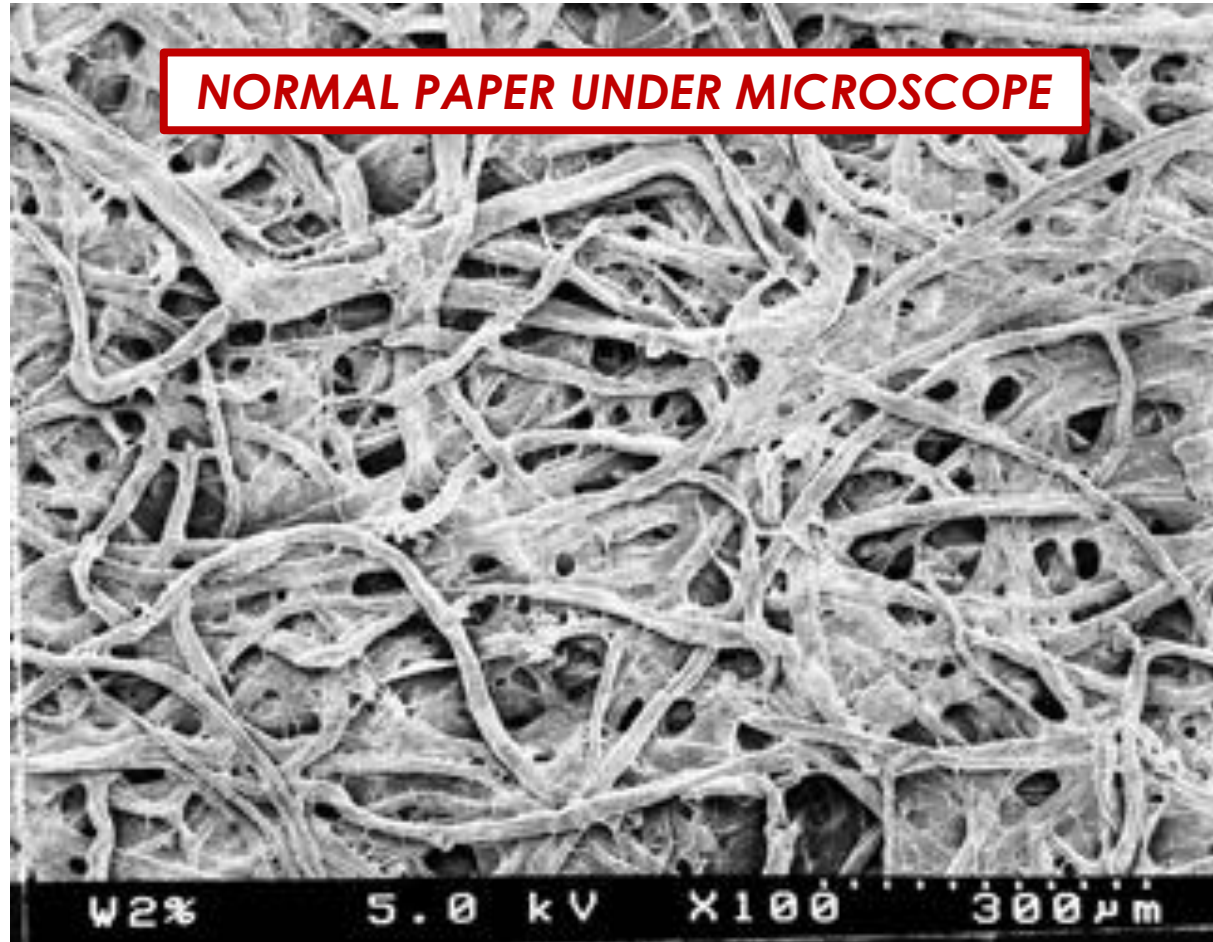
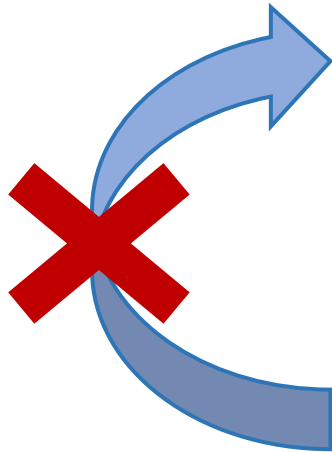
adhesion



viscosity

Default curling direction

Organized cellulose fibers



Source: <https://paper.lib.uiowa.edu/future.php>

LITERATURE

What do we find in science papers?

Research paper

EPL 2011

epl A LETTERS JOURNAL EXPLORING
THE FRONTIERS OF PHYSICS

March 2011

EPL, 93 (2011) 54001
doi: 10.1209/0295-5075/93/54001

www.epljournal.org

...When a piece of tracing paper is placed gently on the surface of a bath of water, it rapidly curls up from one edge and rolls up due to the swelling of the side in contact with water.

With time, as the swelling front propagates through the thickness of the paper, the paper gradually uncurls itself and eventually straightens out...

How wet paper curls

E. REYSSAT^{1,2(a)} and L. MAHADEVAN²

¹ *Physique et Mécanique des Milieux Hétérogènes, UMR 7636 du CNRS, ESPCI - 10 rue Vauquelin, 75005 Paris, France, EU*

² *Engineering and Applied Sciences, Wyss Institute, Harvard University - 29 Oxford St, Cambridge, MA 02138, USA*

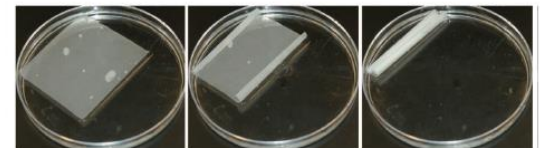
received 12 May 2010; accepted in final form 9 February 2011
published online 10 March 2011

PACS 46.70.De – Beams, plates, and shells
PACS 82.30.Rs – Hydrogen bonding, hydrophilic effects

Abstract – When a piece of tracing paper is placed gently on the surface of a bath of water, it rapidly curls up from one edge and rolls up due to the swelling of the side in contact with water. With time, as the swelling front propagates through the thickness of the paper, the paper gradually uncurls itself and eventually straightens out. We analyze the experimental dynamics of rolling and unrolling of the paper and complement these with a minimal theory that explains the basic observations. Our study might be useful in the context of designing biomimetic devices that work as actuators or harness energy from humidity variations.

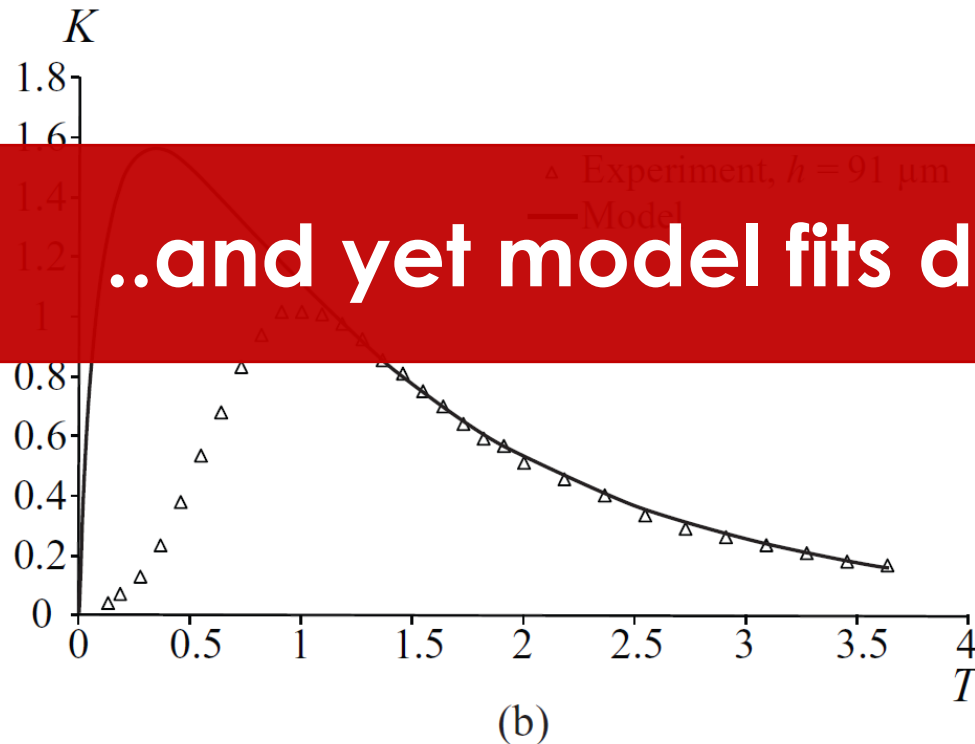
Copyright © EPLA, 2011

Introduction. – From pine and spruce cones [1–3] to wheat awns [4], bimetallic thermostats [5] and micro-actuators [6], many natural or man-made slender systems rely on differential expansion of a layered structure for actuation, driven by variations in humidity or temperature. In the world of artificial materials, few surpass the



Molecular diffusion model

From „How wet paper curls?“



Assumptions and parameters:

1. Normalization of data
2. $D \sim 10^{-10} m^2 s^{-1}$

The diffusion coefficient
3. mass fraction of water saturates at $\phi_S = 0.5$
4. elastic modulus of wet paper

decreases exponentially with water content as:

$$E = E_0 e^{-6.7\phi}$$

5. maximum value of $\varepsilon_\phi(\phi_S)$ between 6 and 10%

Research paper

Timoshenko's bimetallic thermostats 1925

ANALYSIS OF BI-METAL THERMOSTATS

By S. TIMOSHENKO

INTRODUCTION

The following investigation contains a general theory of bending of a bi-metal strip submitted to a uniform heating. This theory is applied in analysis of operation of a bi-metal strip thermostat. The equations are obtained for calculating the temperature of buckling, the complete travel during buckling, and the temperature of buckling in a backward direction.. By using these equations the dimensions of the thermostat for a given temperature of operation and a given complete range of temperature can be calculated. The results obtained are based on certain ideal conditions. For example, it was assumed that the difference in the coefficients of expansion remained constant during heating, that the friction at the supports could be neglected and that the width

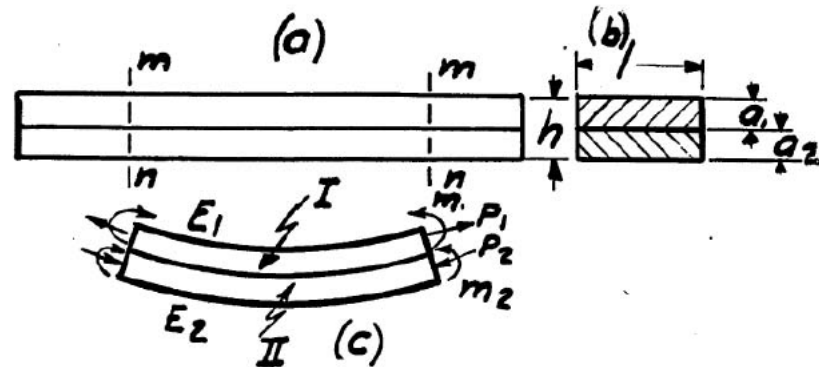
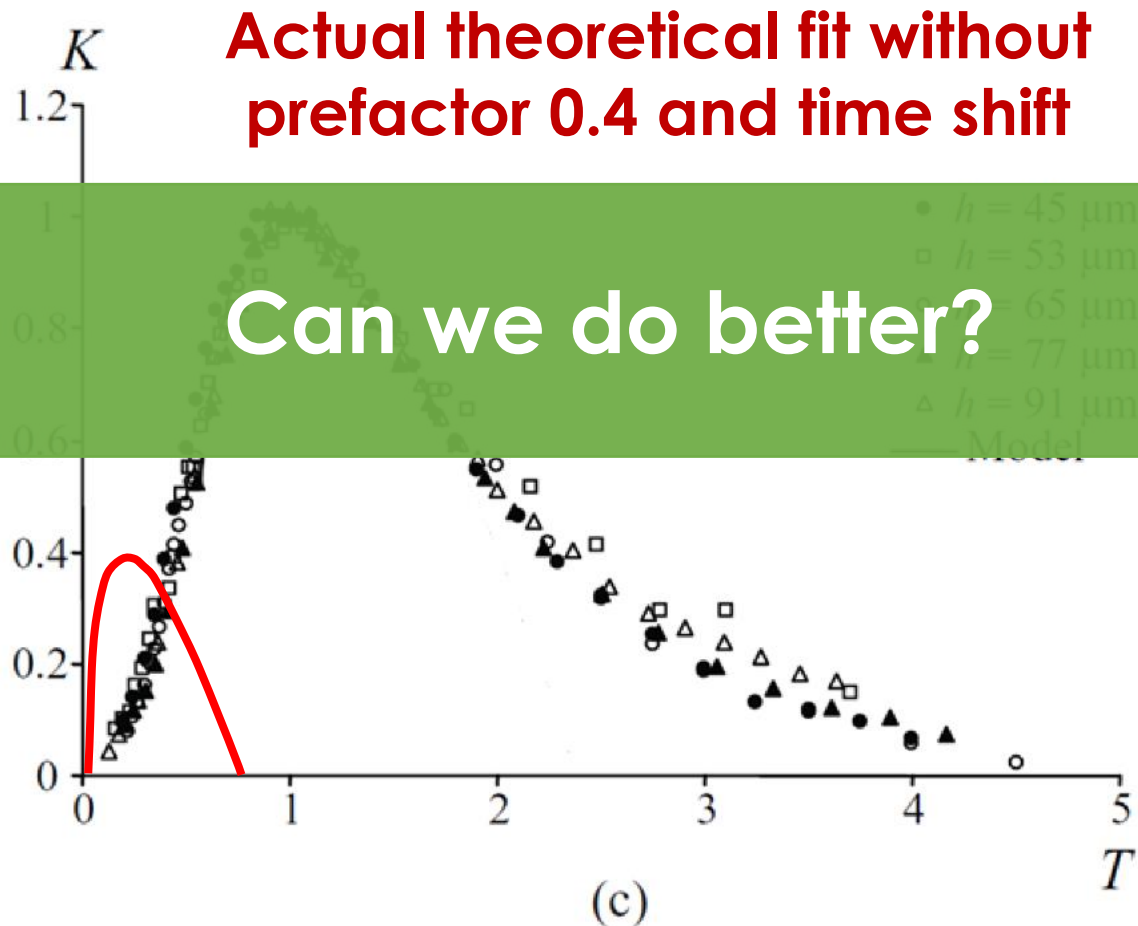


FIG. 1. Deflection of a bi-metal strip while uniformly heated.

of the strip could be considered as being very small. It is the opinion that these assumptions are not far removed from real conditions and the results obtained by using this theory can be considered as a useful guide in choosing the dimensions of thermostats and in discussing the stresses occurring in bi-metallic strips during operation.

Capillary forces model

Based on Timoshenko's "*Analysis of Bi-metallic thermostats*"



EXPERIMENTS

Investigation of relevant parameters

Experimental apparatus

dark
background

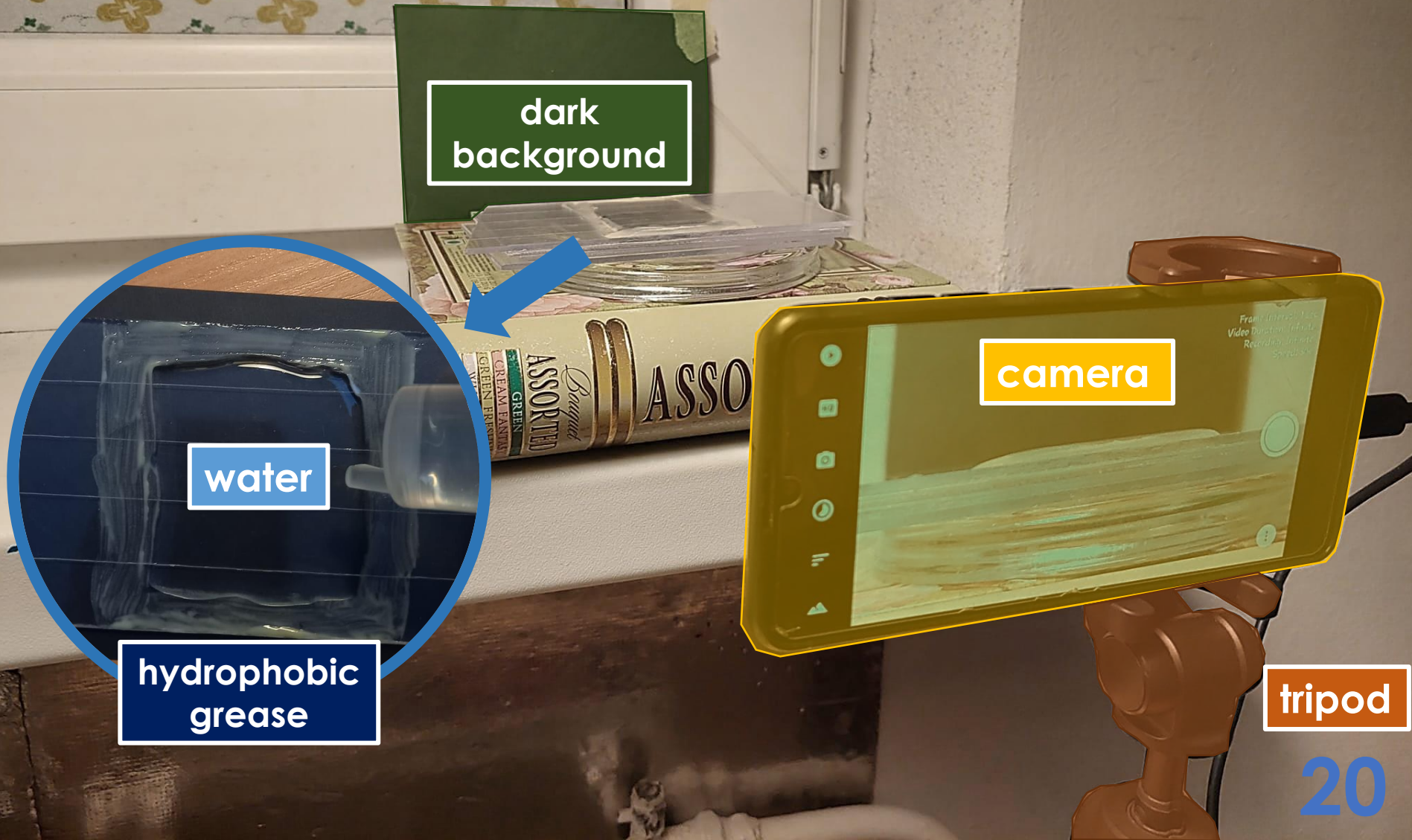
water

hydrophobic
grease

camera

tripod

20



Experimental apparatus

Grooves on
tweezers for
precision



Tracing papers

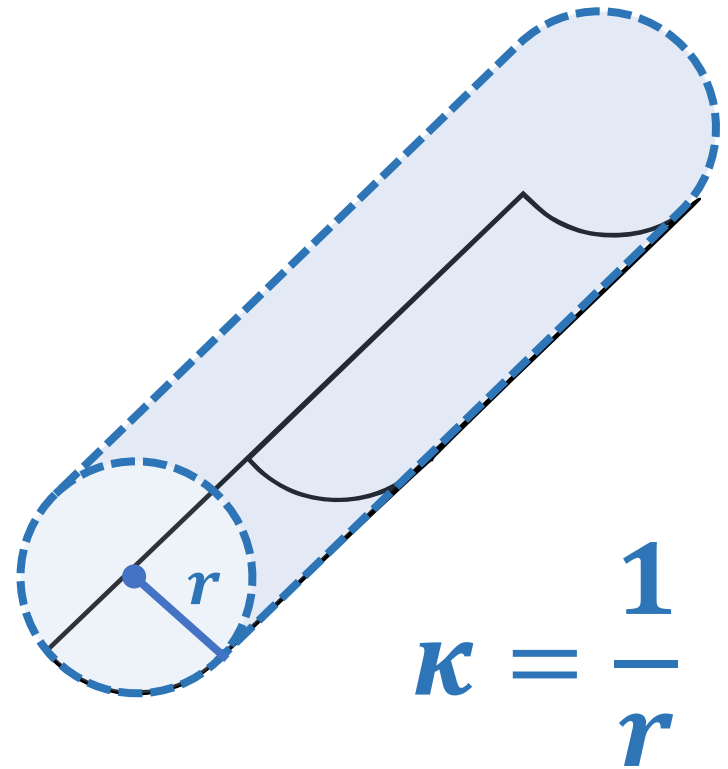
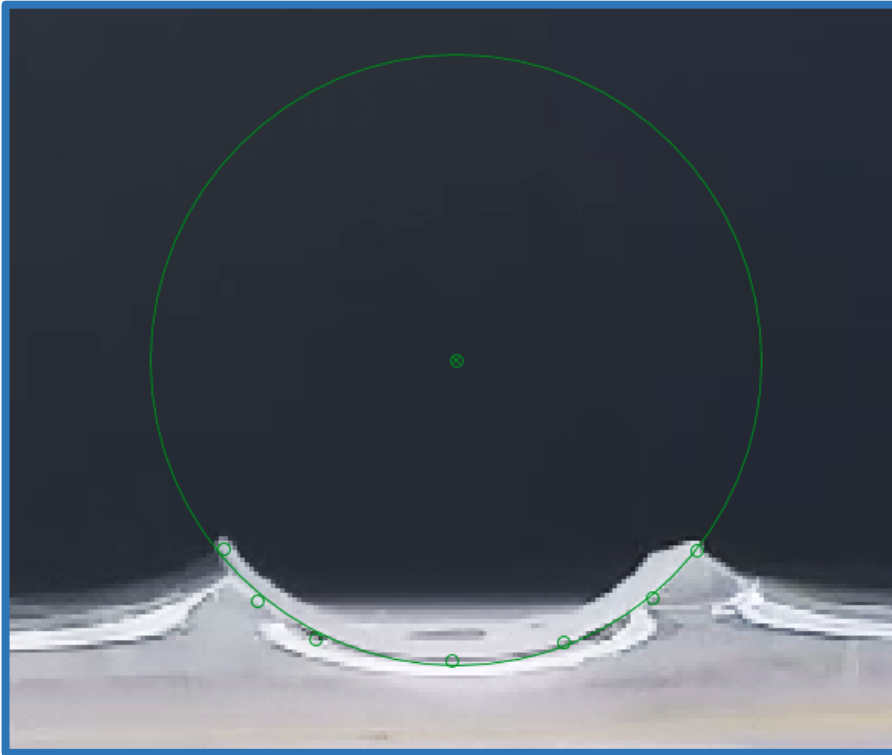


3 types of paper

Grammage (g/m^2)	Thickness (μm)
75	70
90	85
160	130

Methodology

Radius of curvature



Error analysis

Sources of experimental imperfection

**Manufacturing
of paper**

Laying on water

Tracking

**Each measurement
5-time repeated**

n – number of observations
 x – observed value
 $\bar{x}$ – arithmetic mean of data set

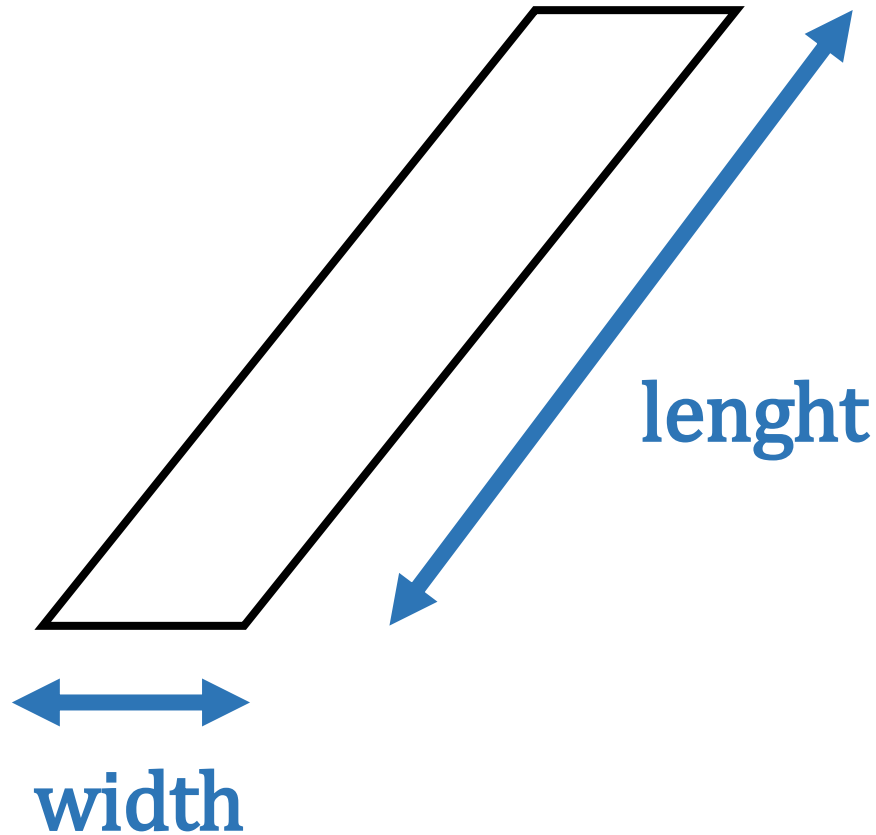
$$\frac{1}{n} \sum |x - \bar{x}|$$

**Average absolute
deviation**

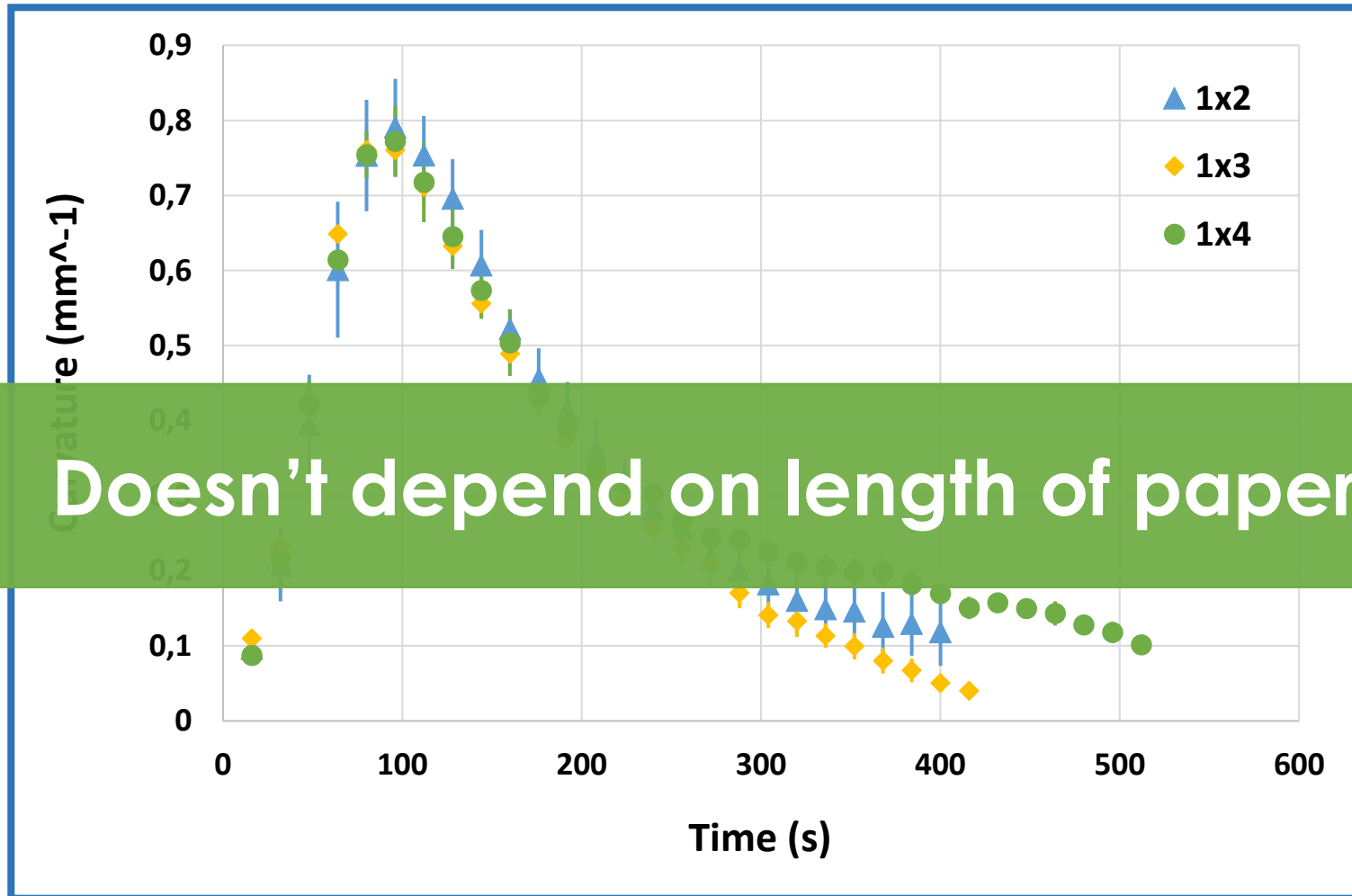
Terminology

Paper dimensions

$W \times L$



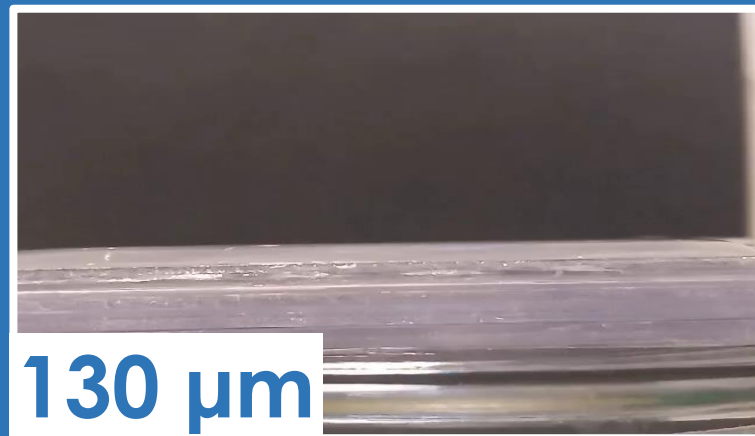
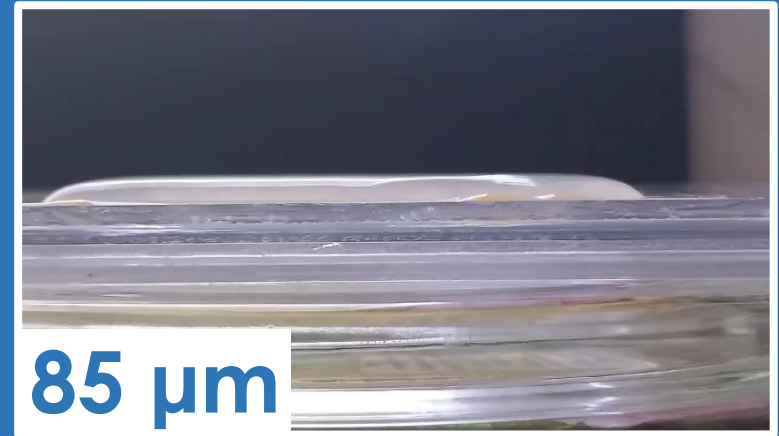
Length effect - 130 μm



Investigating thickness

1x3

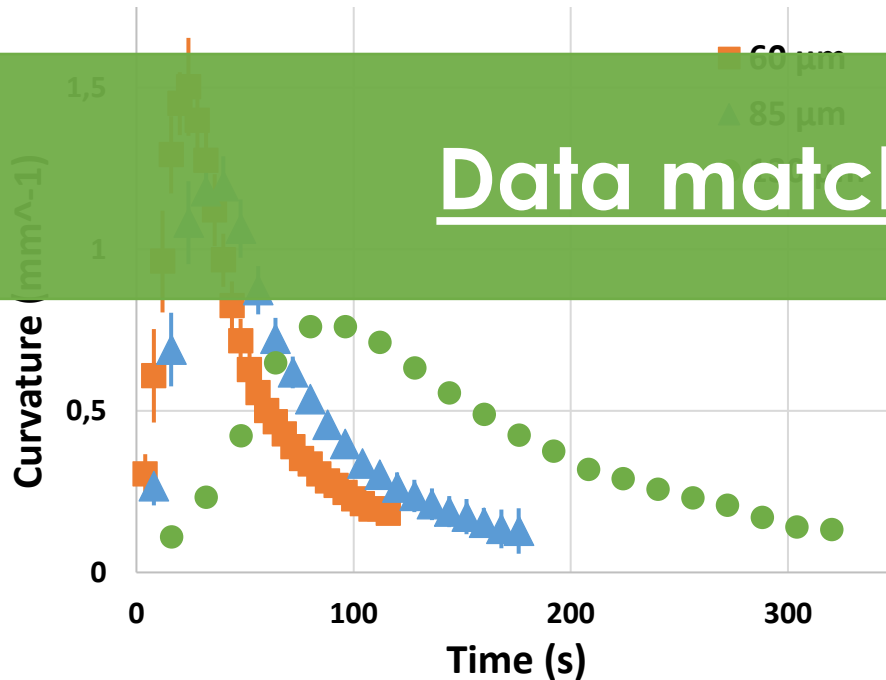
30x
speed



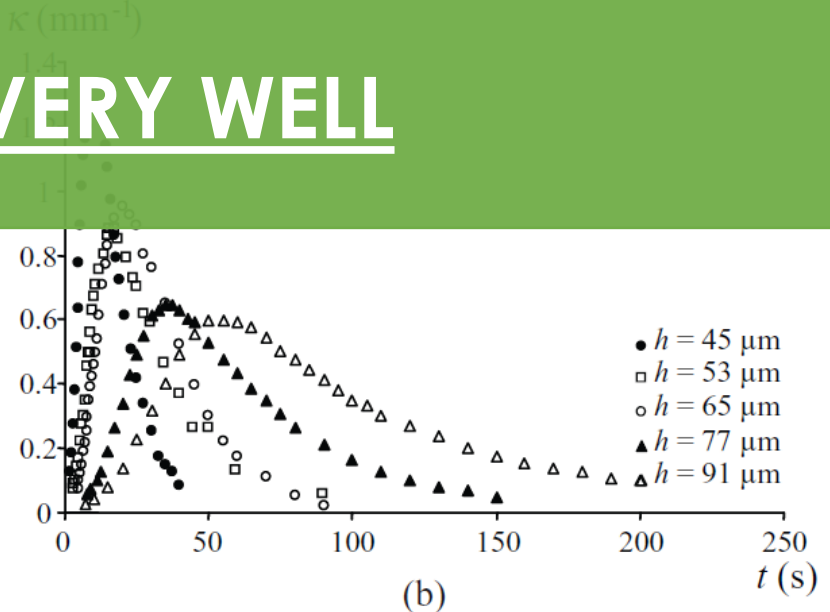
Thickness

Comparison with „How wet paper curls?“

Data match VERY WELL



My data



Data from
„How wet paper
curls?“

Since our experimental
results match results from
papers,
their theory doesn't fit our
data as well.

LET'S MAKE OUR
OWN NEW THEORY!

OUR THEORY

Comparison with experiments

Theory from *„How wet paper curls?“*

What is the transfer mechanism?

To further discriminate between the two possible transport mechanisms, we changed the surface tension γ of the water by adding surfactants. While the molecular diffusion process should be independent of γ , the rate of capillary imbibition should change. Figure 3(a) shows

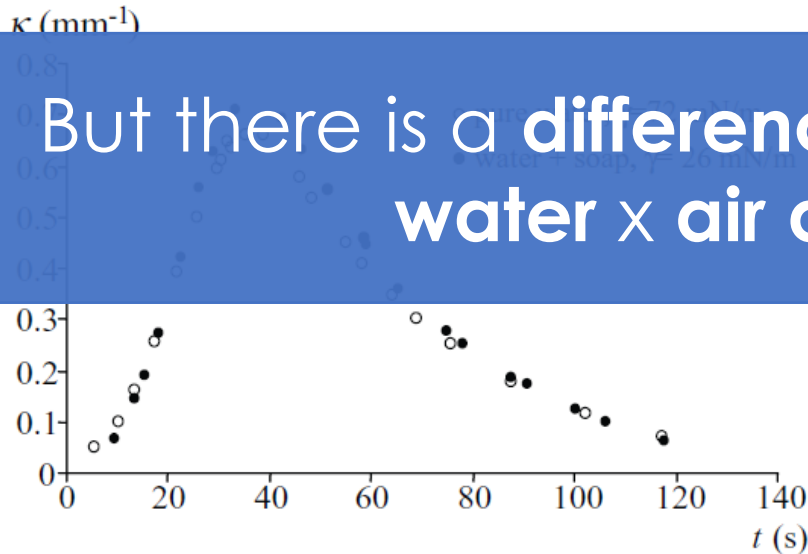
Molecular diffusion?

that for a drop of surface tension from 72 to 26 mN/m, the bending dynamics of a piece of tracing paper is unchanged, while Washburn's law would imply that the process should be 1.5 times slower. All these point to molecular diffusion being the primary mechanism by which water moves through these narrow pores.

What is the transport mechanism?

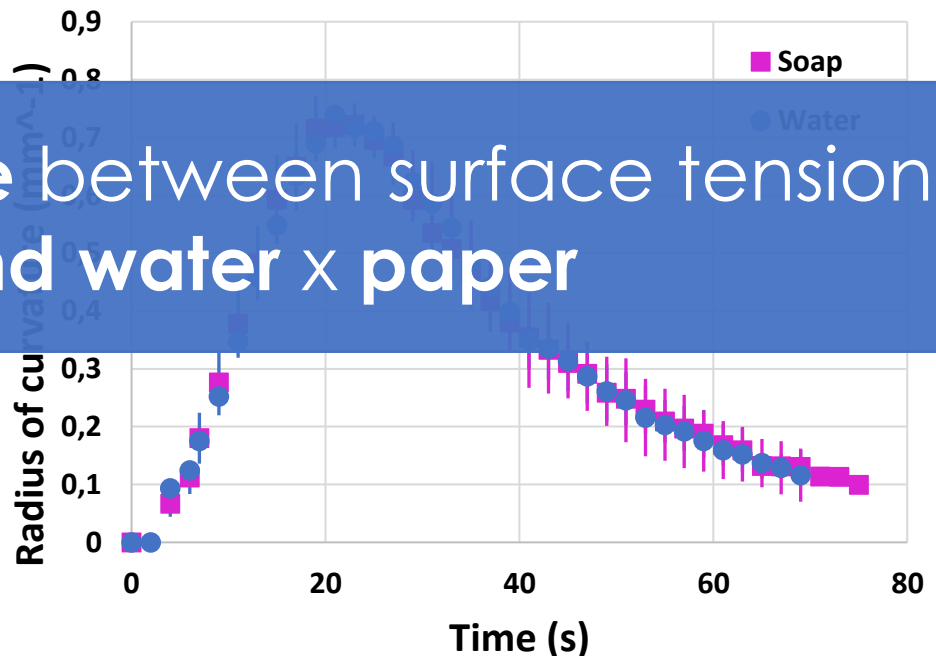
Replicating the experiment with surface tension

But there is a **difference** between surface tension
water x air and water x paper



(a)

From paper



My data

Surface tension change

Redesigning the experiment

We can see clear borderline,
disproving molecular diffusion
as transport mechanism



Water

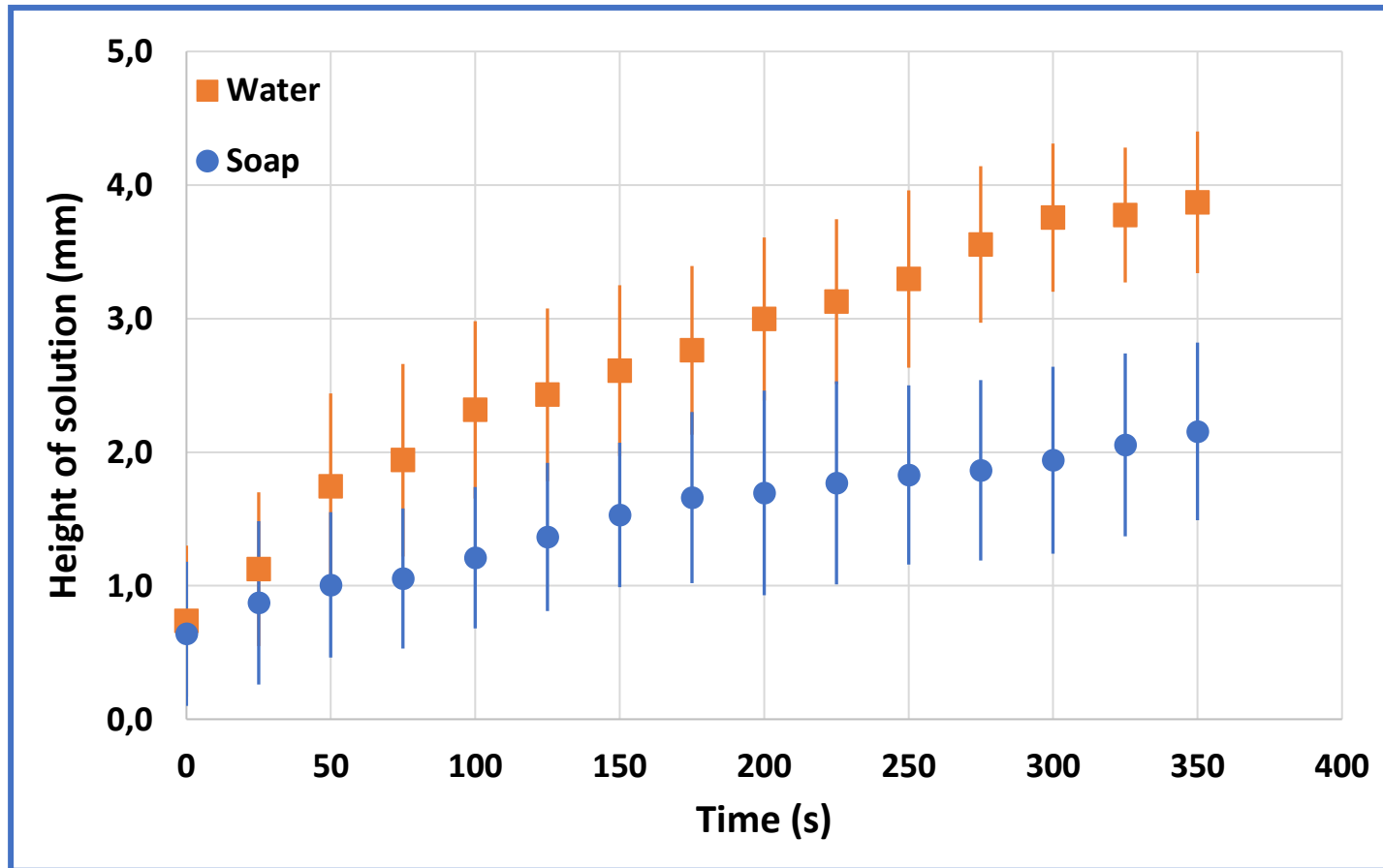
60x
speed



Soap

Surface tension change

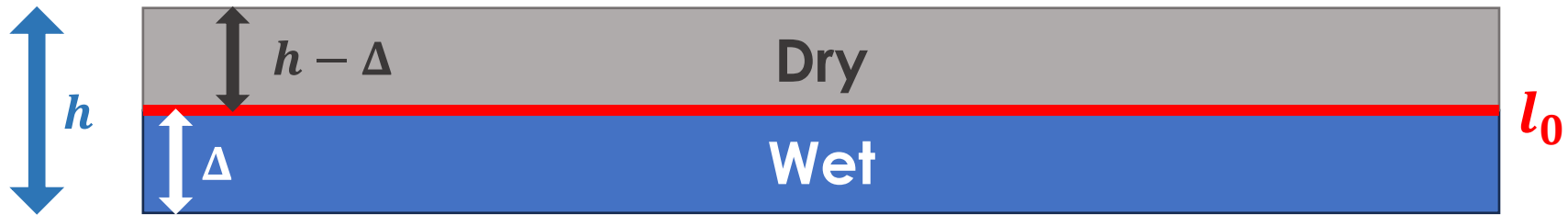
Redesigning the experiment



OUR MODEL

2-layer paper

Same length on connection



l_0 = bottom part of dry layer = top part of wet layer

$$l_0 \left(1 + \epsilon_D + \frac{h - \Delta}{2R} \right) = l_0 \left(1 + f - \epsilon_W - \frac{\Delta}{2R} \right)$$

Change in size due to **stretching and compression**

Change in size due to **curving**

Change in size due to paper **swelling** by water

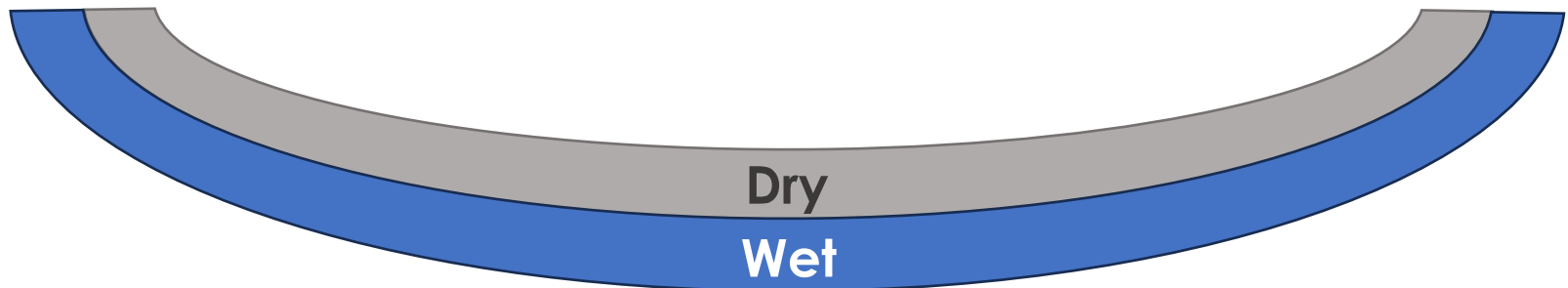
1st order
approximation

Our model

Analytical solution

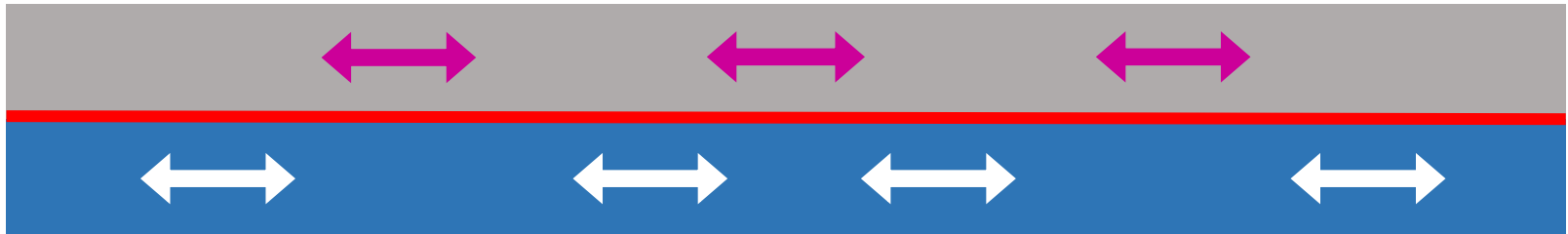
2-layer model with **4** acting **forces**:

1. Stretching of **dry** paper layer



Stretching of dry paper layer

Dry layer stretches » energy used



Paper swells and enlarges

Detailed
derivation in
the appendix

$$E_D = \frac{1}{2} Y_D a(h - \Delta) \left(f - \varepsilon_W - \frac{h\kappa}{2} \right)^2 l$$

Young's modulus of
elasticity for dry paper

=

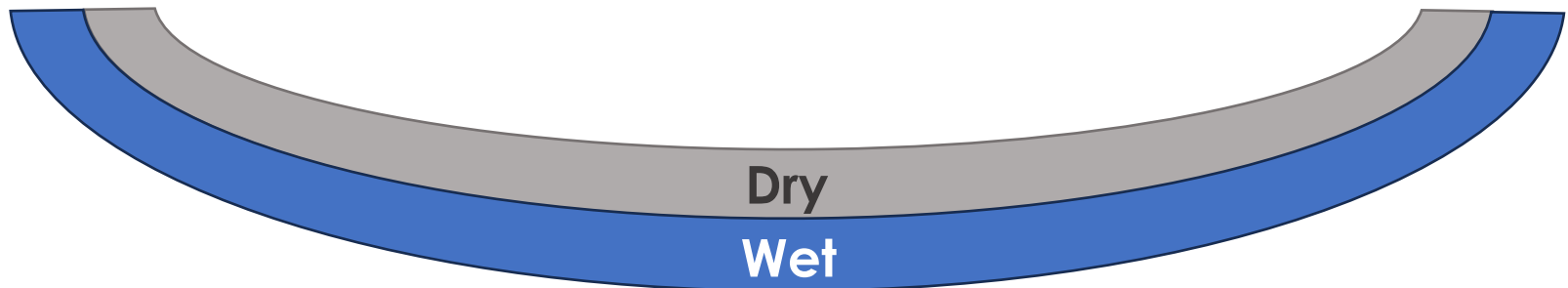
ε_D

Our model

Analytical solution

2-layer model with **4** acting **forces**:

1. **Stretching** of **dry** paper layer
2. **Compression** of **wet** paper layer



Compression of wet paper layer



Paper swells and enlarges

Dry layer compresses » energy used

Detailed
derivation in
the appendix

$$E_W = \frac{1}{2} Y_W a \Delta \varepsilon_W^2 l$$

Young's modulus of
elasticity for wet paper

Curving energy

Derived from cantilever beam bending equation

Detailed
derivation in
the appendix

Curving of
dry layer

$$E_{DC} = \frac{1}{32} Y_D a (h - \Delta)^3 l \kappa^2$$

Curving of
wet layer

$$E_{WC} = \frac{1}{32} Y_W a \Delta^3 l \kappa^2$$

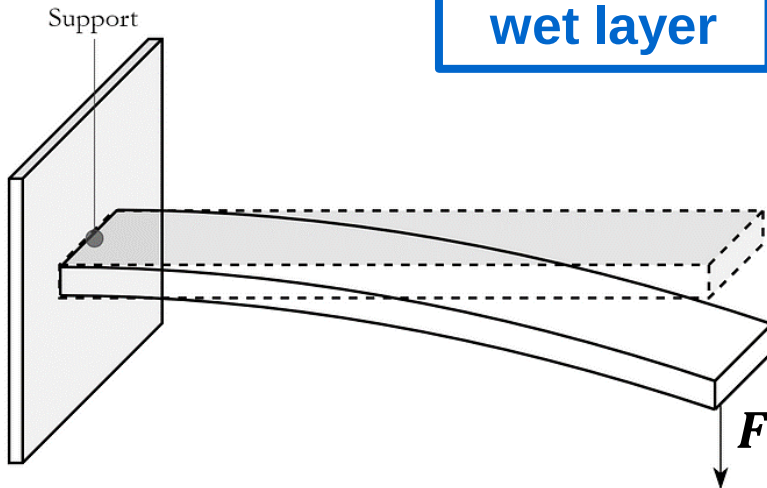


Image created on draw.io by Anupsa Swain

Cantilever beam bending equation:

$$y = \frac{4Fl^3}{Yab^3}$$

Our model

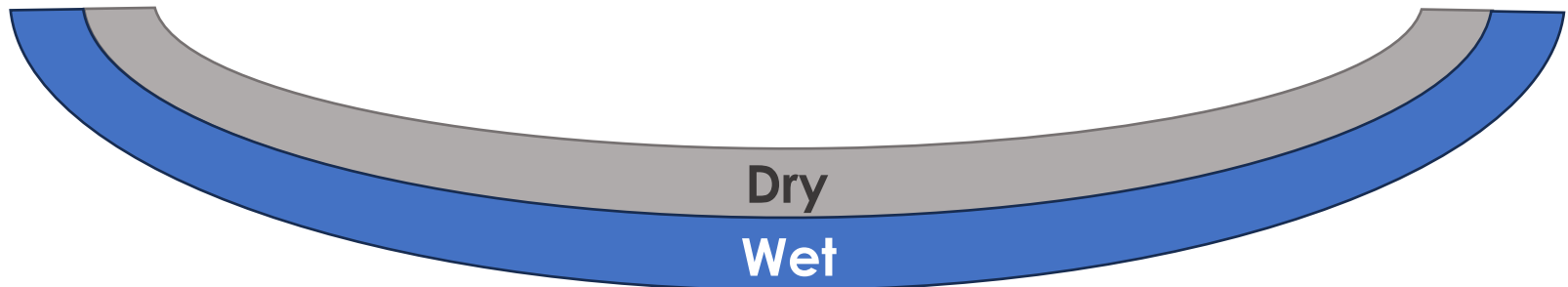
Analytical solution

2-layer model with **4** acting **forces**:

1. Stretching of dry paper layer
2. Compression of wet paper layer
We find κ and Δ dependency!

3. Curvature of dry paper layer

4. Curvature of wet paper layer



Minimalization of energy

Derivative of sum of energies

$$E_D = \frac{1}{2} Y_D a(h - \Delta) \left(f - \varepsilon_W - \frac{h\kappa}{2} \right)^2 l$$

+

$$E_W = \frac{1}{2} Y_W a \Delta \varepsilon_W^2 l$$

+

$$E_{DC} = \frac{1}{32} Y_D a(h - \Delta)^3 l \kappa^2$$

+

$$E_{WC} = \frac{1}{32} Y_W a \Delta^3 l \kappa^2$$

$$= E_{total}$$

Detailed
differentiation
in the appendix

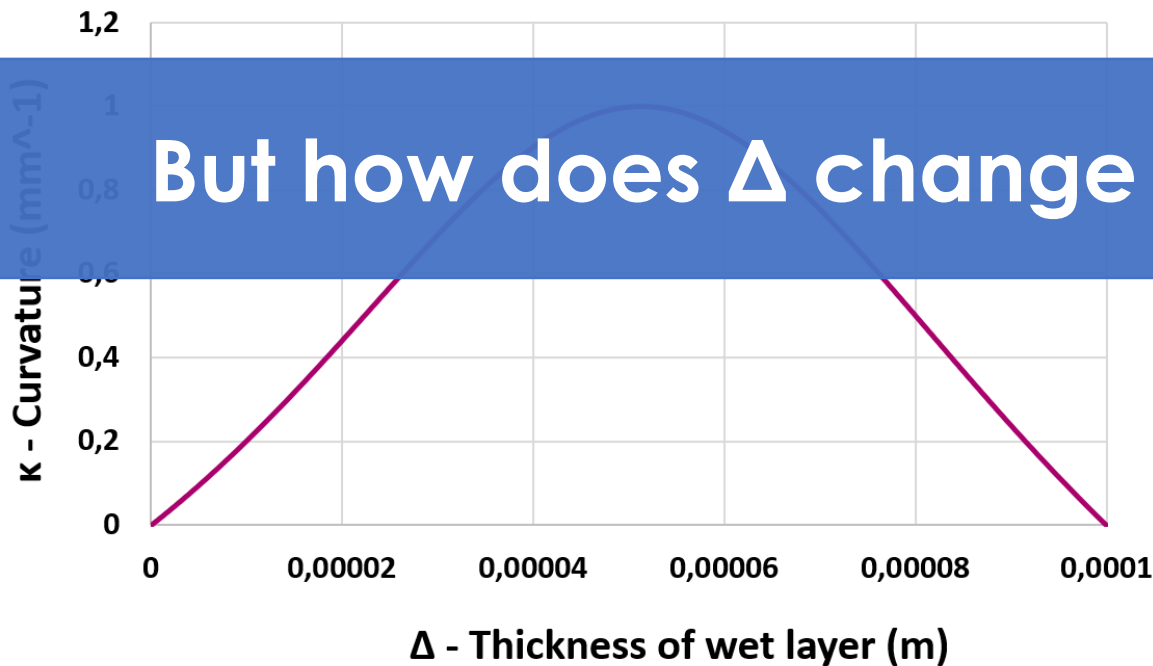
$$\frac{dE}{d\varepsilon_W} = 0 \rightarrow \text{minimum}$$

Substitute ε_W
as constant

$$\frac{d}{d\kappa} E_{total} = 0$$

Δ and κ relationship

$$\kappa = \frac{8\Delta fh(-\Delta + h)Y}{h^4 + \Delta^2 h^2(6 - 7Y) + 4\Delta h^3(-1 + Y) + \Delta^3 h(-4 + 5Y) + \Delta^4(-1 + Y)^2}$$



But how does Δ change in time?

Parameters:
 $h = 1 \times 10^{-4} \text{ m}$
 $f = 0.1$
 $Y = 0.9$

Capillary rise of liquid

Hagen-Poiseuille law

$$\Delta p = \frac{8\pi\eta h Q}{\pi R^4}$$

$$\frac{2\sigma \cos \theta}{r} - \underbrace{\rho g \Delta}_{\text{negligible}} = \frac{8\eta \Delta \dot{\Delta}}{r^2}$$

$$\underbrace{\frac{2\sigma \cos \theta}{r}}_{\text{capillary force}} - \underbrace{\frac{8\eta \Delta \dot{\Delta}}{r^2}}_{\text{friction force}} = \rho \dot{\Delta}^2$$

Non-linear
differential equation
with
no analytic solution

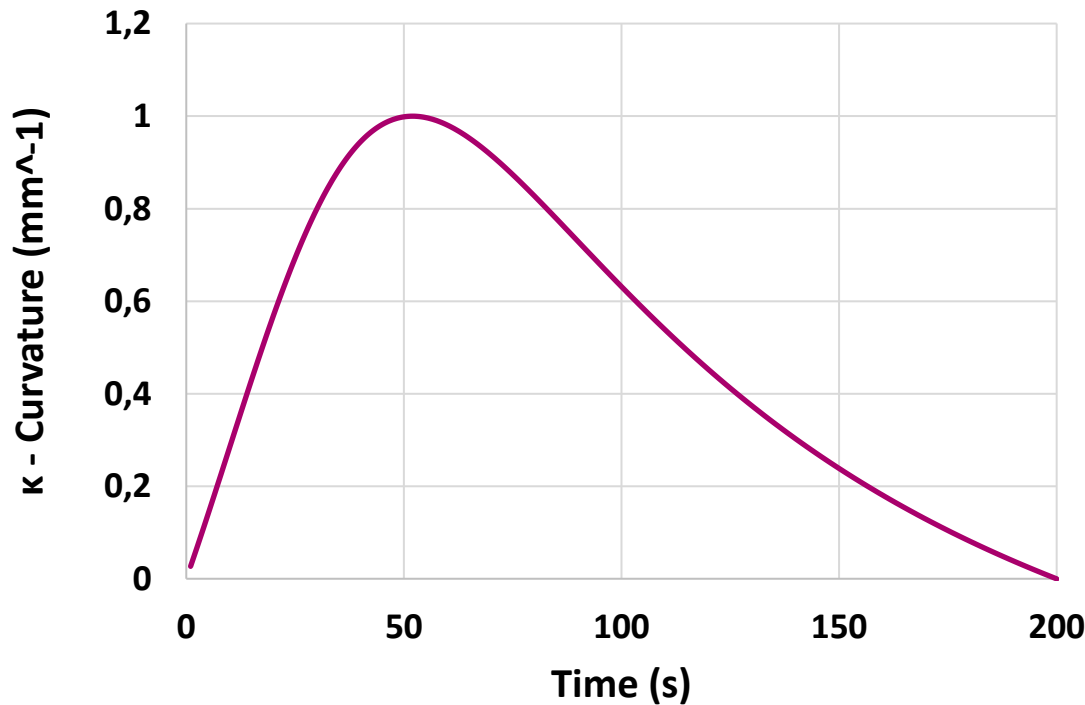
We decided to use empirical model

Analytical result

Empirical model

$$\Delta = \frac{v_0}{(1 + qt)} t$$

$$\kappa = \kappa(\Delta(t))$$



Parameters:

$$h = 1 \times 10^{-4} \text{ m}$$

$$f = 0.1$$

$$Y = 0.9$$

$$v_0 = 1.5 \times 10^{-6} \text{ m/s}$$

$$q = 0.01$$

Comparison

Between theory and experiments

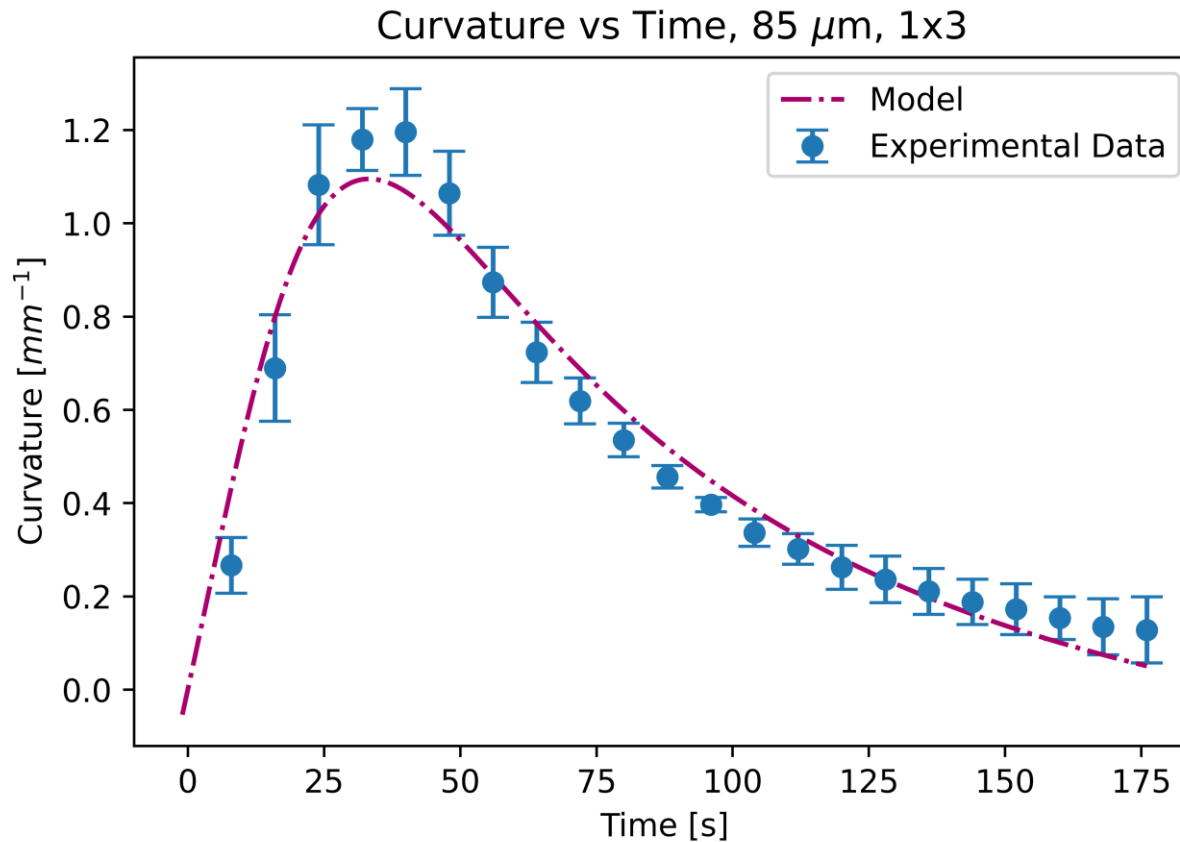
Parameters

Guesses and Boundaries

		Parameter	Symbol	Guess	Boundaries
Material properties	{	Thickness of paper	h	measured	fixed
		Expansion of wet paper due to water	f	0.08	*0.06 – 0.1
		Ratio of Young's moduli	$Y = \frac{Y_W}{Y_D}$	0.6	*0.2 – 1
		Initial velocity of water sucked up into capillaries	v_0	$\frac{h}{\text{peak time}}$	$\pm 50\%$
		Damping of water flow	q	$\frac{1}{\text{peak time}}$	free

*From E. Reyssat and L. Mahadevan: „How wet paper curls?“; 2011; EPL Journal

Theory vs Experiments



Parameters:

$$h = 8.5 \times 10^{-5} \text{ m}$$

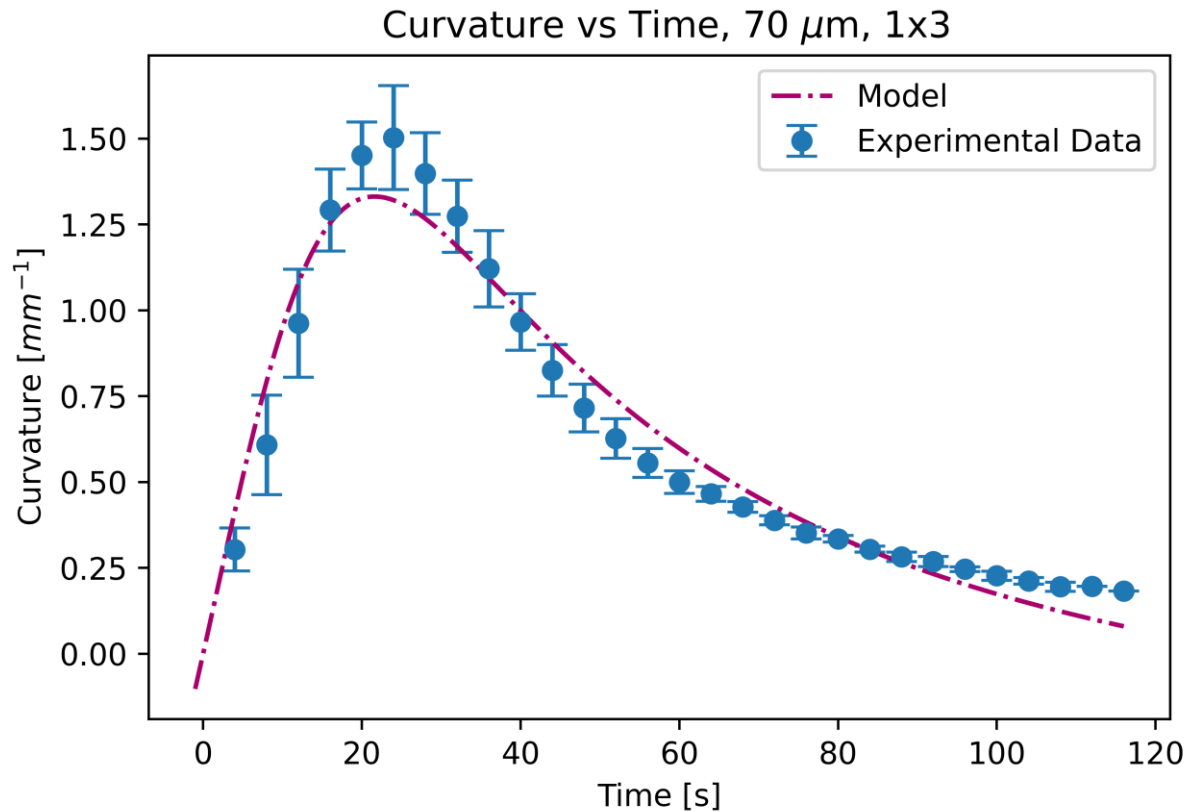
$$f = 0.09$$

$$Y = 0.97$$

$$v_0 = 2.2 \times 10^{-6} \text{ m/s}$$

$$q = 0.02$$

Theory vs Experiments



Parameters:

$$h = 7 \times 10^{-5} \text{ m}$$

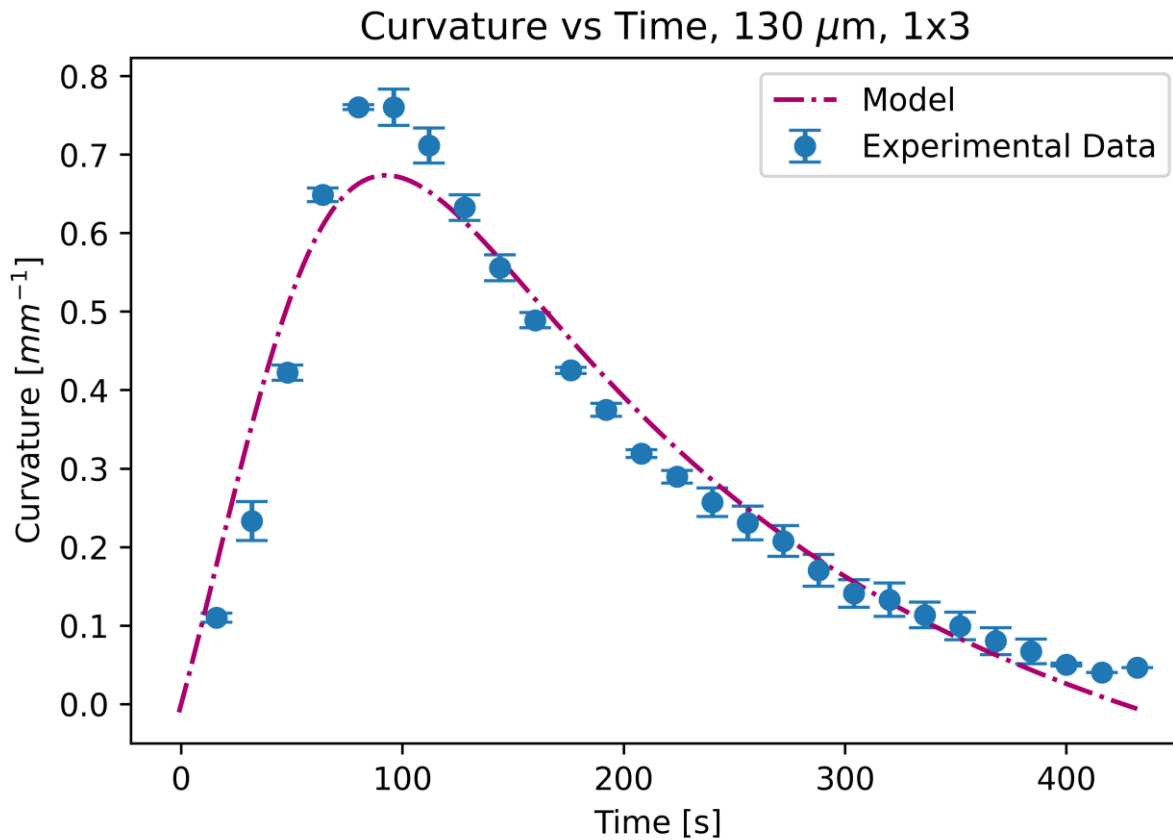
$$f = 0.08$$

$$Y = 0.99$$

$$v_0 = 2.3 \times 10^{-6} \text{ m/s}$$

$$q = 0.03$$

Theory vs Experiments



Parameters:

$$h = 1.3 \times 10^{-4} \text{ m}$$

$$f = 0.09$$

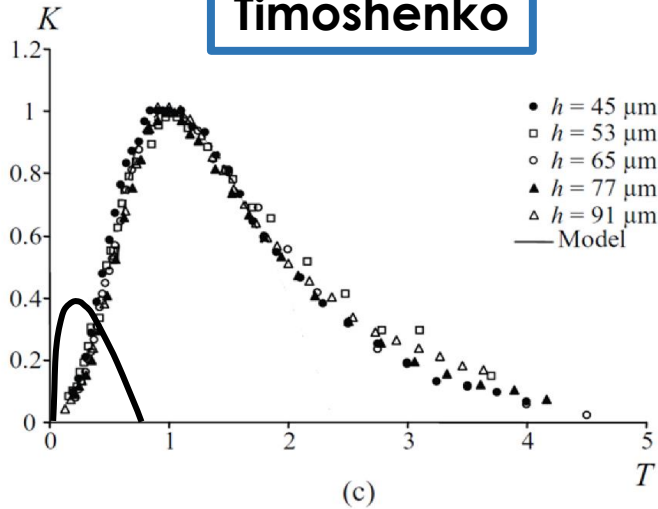
$$Y = 0.84$$

$$v_0 = 1.2 \times 10^{-6} \text{ m/s}$$

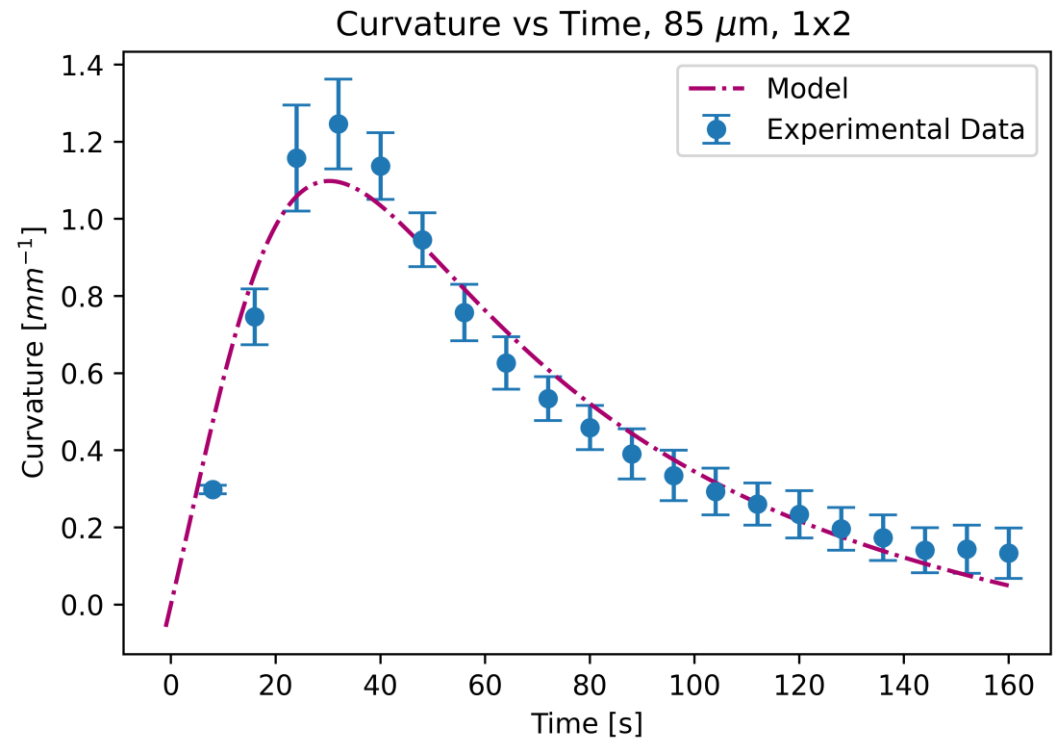
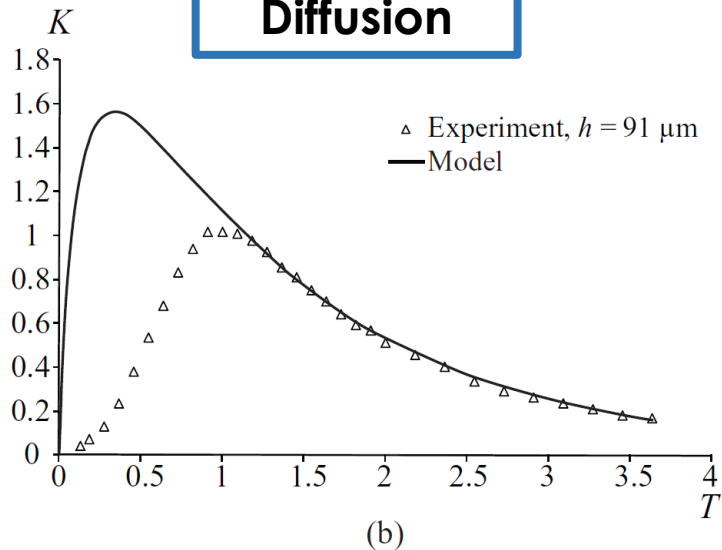
$$q = 0.01$$

Conclusion

Timoshenko



Diffusion



Our model

THANK YOU

APPENDIX

Why empirical?

Hagen-Poiseuille law

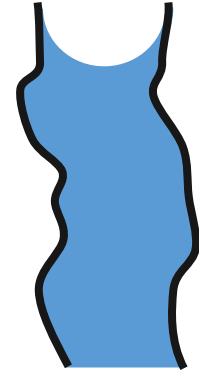
$$\Delta p = \frac{8\pi\eta h Q}{\pi r^4}$$

$$\frac{2\sigma \cos \theta}{r} - \underbrace{\rho g \Delta}_{\text{negligible}} = \frac{8\eta \Delta \dot{\Delta}}{r^2}$$

$$\underbrace{\frac{2\sigma \cos \theta}{r}}_{\text{capillary force}} - \underbrace{\frac{8\eta \Delta \dot{\Delta}}{r^2}}_{\text{friction force}} = \rho \dot{\Delta}^2$$

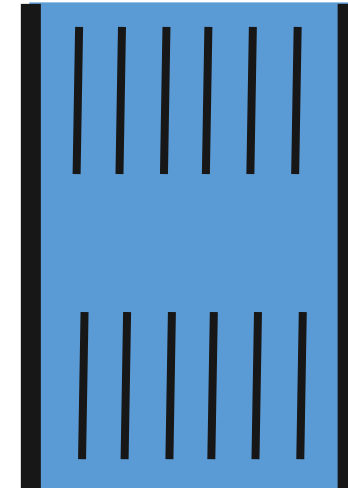


In reality



Simple capillary

Complex capillary



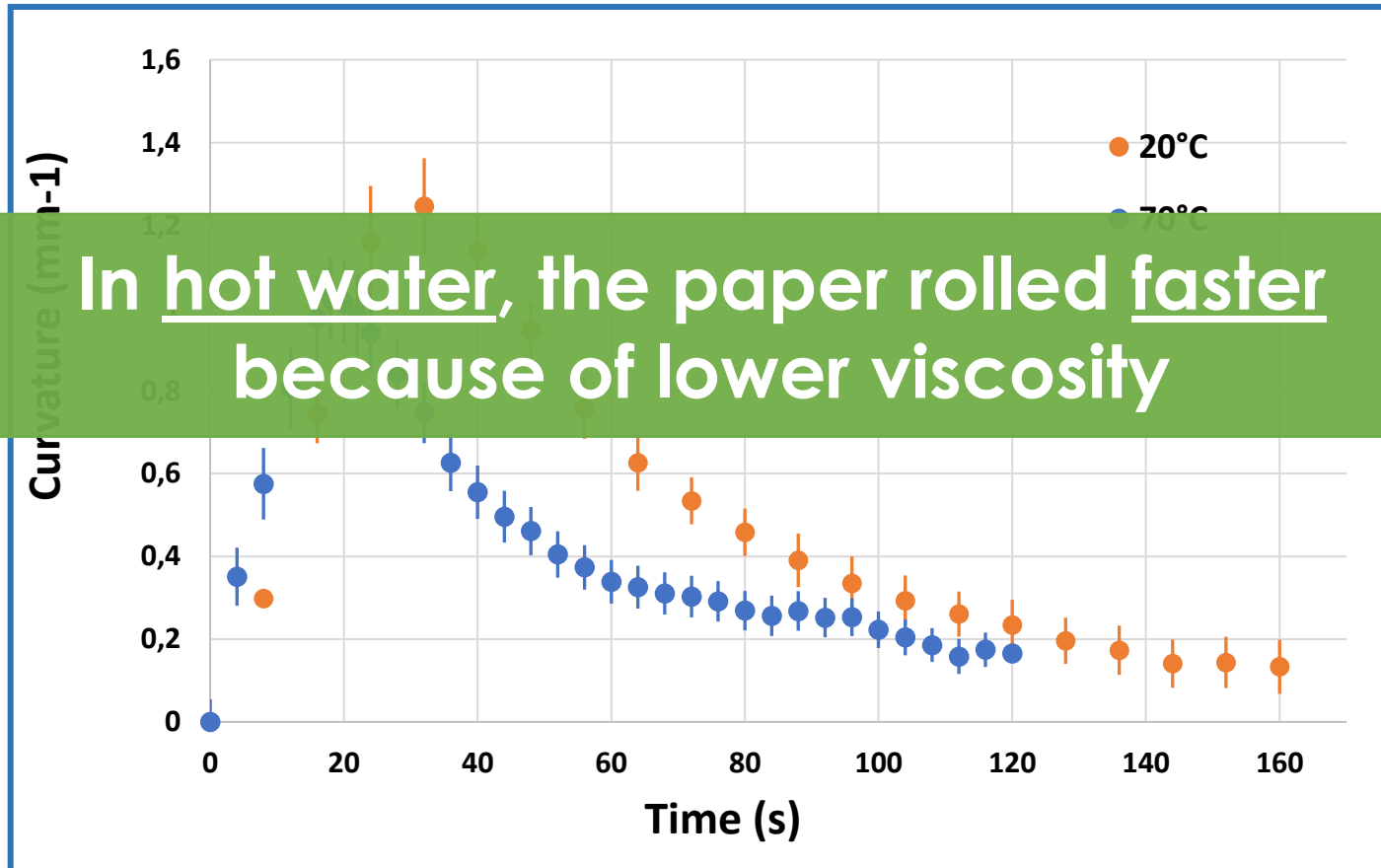
Temperature

3D printer heating bed



Temperature

85 μm



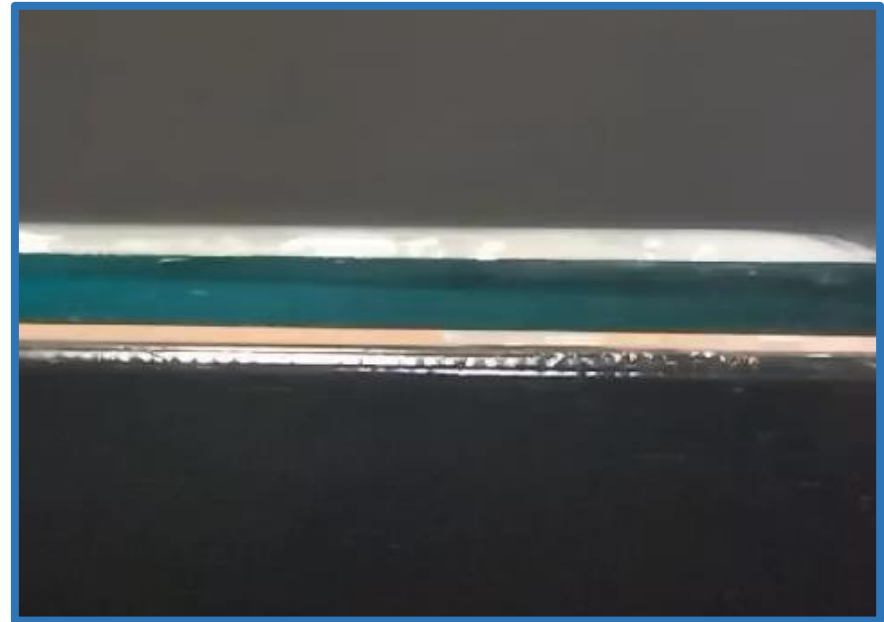
Rolling

15x
speed

Exploring width



30x
speed



Attached
on one
end